TASK 6

Dataset Preparation and Preprocessing

Face Monitoring Model and Phone, Drink & Food Detector (YOLO26n), version 2

Project	Real-Time Driver Monitoring System (RTDMS)
Scope	Face monitoring (MediaPipe + EAR) and phone, drink and food detection (YOLO26 nano)
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Programme	B.Sc. Graduation Project, Computer Engineering, AASTMT

Contents

1. Introduction 3
1.1 Summary at a glance 3
1.2 What changed since version 1 3
2. Face Monitoring Model 3
2.1 Datasets 4
2.2 Splitting 4
2.3 Preprocessing 5
3. Phone, Drink and Food Detector (YOLO26n) 6
3.1 How it works 6
3.2 Datasets we used 6
3.3 Organising the data 7
3.4 Labelling 7
3.5 Cleaning 7
3.6 Preprocessing 8
3.7 Splitting 8
3.8 Size and class balance 9
4. Progress: How It Went 10
5. Data Quality and Ethics 10
6. Assumptions and Limitations 11
7. Next Steps 11
8. References 11

1. Introduction

Our driver monitoring system uses two camera models:

- **Face model**: watches the eyes and head. It detects **drowsiness** and **looking away**.
- **Object detector**: finds a **phone**, a **drink** or **food** in the picture. It detects **phone use**, **drinking** and **eating**.

This report shows which data each model uses, and how we **organised**, **labelled**, **cleaned**, **prepared** and **split** it.

1.1 Summary at a glance

Table 1: *The models and their data*

Model	What it detects	Training data	Test data
Face model (MediaPipe + EAR)	Eye closure, blinks, PERCLOS [1], head pose, gaze	None needed (Google's ready model [2])	CEW, 2,423 photos [3]
Eye-state CNN (comparison)	Eyes open or closed	MRL Eye, 84,898 IR images [4]	8 MRL people never seen
Object detector (YOLO26n) [5]	Phone, drink and food boxes	5 sources, 53,889 images	10 new SynDD drivers + 2 unseen datasets

1.2 What changed since version 1

> **Why we changed the phone model**
>
> Version 1 used a ready **image classifier** trained on **State Farm** [16]: colour photos taken from the **passenger side**.
>
> Our camera is **infrared** and sits at the **centre mirror**, so it sees the driver from a different angle.
>
> We replaced it with an **object detector**. It draws a **box** around the phone, cup or food, so the system also knows **where** it is (for example, a cup **at the mouth**).

2. Face Monitoring Model

This part is **unchanged** since version 1. The face model finds **478 points** on the face [2]. From them it measures how open the eyes are and where the head points. Simple rules then decide the driver state. It uses Google's **pre-trained** models, so it needs **no training data**; we use data only to **test** it.



Figure 1: *Face model, from camera frame to driver state*

2.1 Datasets

Table 2: *Face datasets*

Dataset	Size	Image type	Labels	Used for
CEW [3]	2,423 face photos (1,192 closed, 1,231 open)	Colour, 100 × 100 faces	Open / closed (folder name)	Choosing thresholds and testing
MRL Eye [4]	84,898 eye images from 37 people	Infrared, grey eyes	8 labels in each file name (eye state, glasses, light...)	Training the eye-state CNN

2.2 Splitting

CEW was split into a **tuning set (30%)**, used only to choose thresholds, and a **test set (70%)**, used only for the final score.

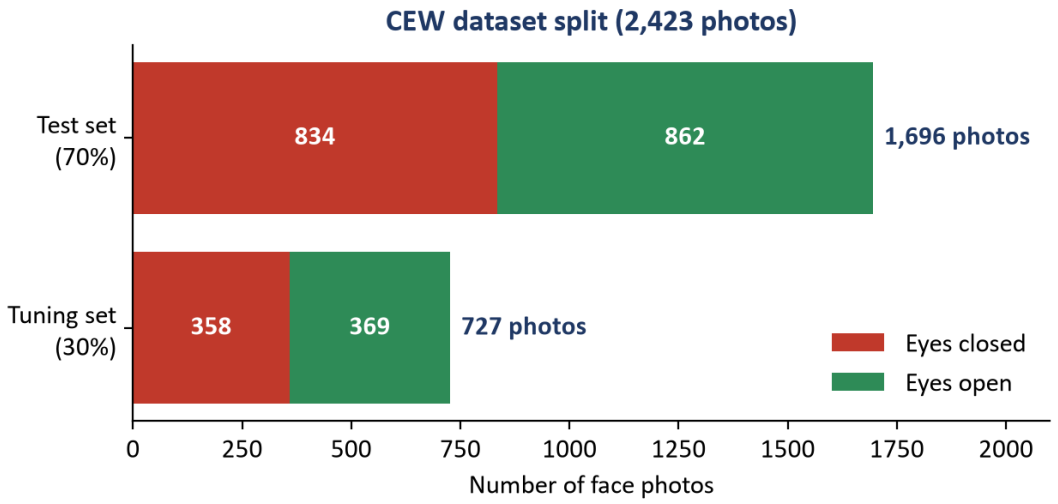


Figure 2: *CEW split into tuning and test sets*

MRL was split **by person**: nobody in the test set appears in training.

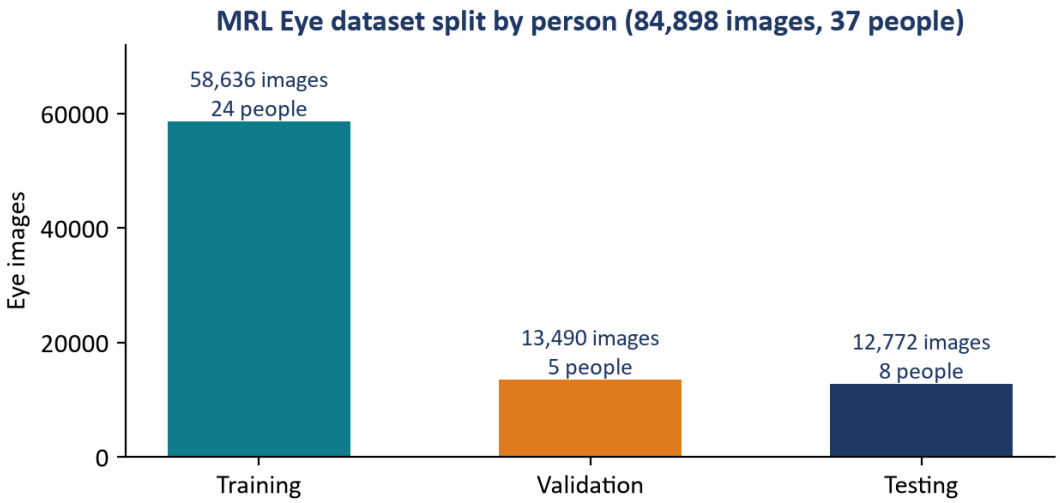


Figure 3: *MRL split by person*

Why split by person?

If the same person is in training and testing, the model can **memorise that person**, and the score looks better than it really is. Splitting by person gives an **honest score for new drivers**. We use the same rule for every dataset in this report.

2.3 Preprocessing

Table 3: *Preprocessing of the face data*

Step	Live camera frames	CEW photos (test)	MRL eyes (training)
Size	640 × 480, 30 frames/s	50% border added, then 400 × 400	32 × 32
Colour	RGB; at night grey → 3 channels + CLAHE contrast [14]	RGB	Grey
Normalisation	Inside MediaPipe; EAR [15] scaled to each driver's calibration	Inside each model	Per image: minus mean, divided by contrast
Cleaning	3-frame median filter on EAR	-	-
Augmentation	None	None	9 types (flip, rotate ±15°, zoom, shift, light, blur, noise...)

3. Phone, Drink and Food Detector (YOLO26n)

3.1 How it works

A small detector (**YOLO26 nano**, 2.5 million weights) looks at the driver and draws boxes around a **phone**, **drink** or **food**. Short rules turn the boxes into alerts.

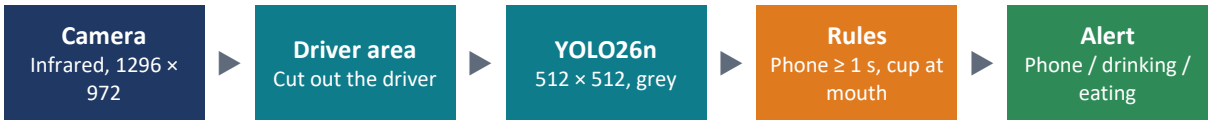


Figure 4: *Phone, drink and food detector*

3.2 Datasets we used

No single public dataset matches our camera, so we combined **five training sources** and kept **two more datasets for testing only**.

Table 4: *Datasets for the object detector*

Dataset	What it is	Size we used	How we used it
SynDD [6]	Acted driving video, **infrared**, camera **at the mirror** (like ours)	69 drivers, 138 videos → 18,318 frames	**Main training data** + test (by driver)
Open Images V7 [7]	Everyday photos with boxes	16,996 photos	Teaches what phones, cups and food look like
DMS_Project [8]	Driver-camera images, IR and colour	10,198 images	Training
ADMS_FINAL_3 [9]	In-car colour images	2,698 images	Training
EBDD [10]	Driver video through the windshield, colour	13 drivers, 2,093 frames	Training + test (by driver)
3MDAD [11]	50 drivers, front camera, 16 actions	650 clips	Test only
DAD [12]	**Real infrared**, front + top camera	3 drivers, 66 min normal driving	Test only

We also tried three more sources. They made results **worse**, so they are **not used**:

Table 5: *Data we tried and dropped*

Data	What happened	Kept?
State Farm [16]	Passenger-side colour photos: wrong camera angle	Dropped
SynDD extra views (dashboard, side window), 20 drivers	False phone alarms rose from 5 to 24 per hour	Not kept
SynDD A2, 15 unlabelled drivers	Labels made by a model were too noisy; more false alarms	Not kept

3.3 Organising the data

Table 6: *Folder layout (on the project drive, never on GitHub)*

Folder	What is inside
raw/	Original downloads, never changed (EBDD, 3MDAD, DAD)
public/syndd/A1/user_id_…/	One folder per driver: the videos + the official label file (.csv)
det_v2/<source>/images/train | val | test	Frames of each source, already split
det_v2/<source>/labels/…	One text file per image with its boxes
det_v2/merged_b/	The **final combined set** used for training

Each box file has one line per object: **class, x, y, width, height** (all as 0–1 of the image size). Classes: **0 = phone, 1 = drink, 2 = food**.

3.4 Labelling

- **SynDD** gives the **start and end time** of every action. We turned its 16 action types into ours: phone call → **phone at ear**, texting → **phone in hand**, **drinking**, **eating**, **normal**. Talking, singing, yawning and hand on head are kept as **look-alikes** (no object).
- **Boxes** in video were drawn **automatically** by a large "teacher" model (YOLO26 medium at 1280 px, plus YOLO-World [13] for food) on a zoomed view of the driver.
- A box is **kept only if it matches the action**: a phone near the ear during a call, a cup at the mouth while drinking.
- **Open Images** and **Roboflow** already have boxes; we only renamed the classes ("using iphone" → phone; "Mug", "Bottle", "Tin can" → drink).
- Checks: label sheets were looked at by eye; **495 test frames** are ready for a hand check (still to do).

3.5 Cleaning

- Action frames where the teacher found **no matching box** were **removed** (the phone was hidden). Keeping them would teach the model "no phone here".
- Normal-driving frames where the teacher saw a **confident phone** were removed (maybe a phone in a holder).
- "Picking up the phone" spans are **ignored** when scoring: they often show a real phone.
- Roboflow boxes covering more than **12%** of the image (8% for drinks) were removed: they marked the whole action, not the object. Images with none of our objects were removed.
- Open Images group boxes and drawings were skipped.
- One SynDD label error was found: a 4-second "phone call" with no phone.

3.6 Preprocessing

1. **Frame extraction**: 1 frame every **0.5 s** during actions and every **1.5 s** during normal driving → **18,318** SynDD frames from 1,653 action clips.
2. **Resizing**: frames stored at most 960 px wide; the model trains and runs at **512 × 512**.
3. **Infrared look**: every training image is turned **grey**; half also get camera effects (blur, noise, JPEG, contrast).
4. **Normalisation**: pixel values scaled to **0–1** inside the model.
5. **Balancing**: at most **4,000** Open Images photos per class; **7,913** extra phone images with a real **hand pasted over 30–60%** of the phone (teaches hidden phones); **7,010** grey copies of colour images.
6. **Augmentation** in training: mosaic, left-right flip, zoom ±50%, shift ±10%, rotate ±5°, brightness ±40%.
7. **Live**: camera frame → cut out the driver area → 512 px → model, about **10 frames per second** on the Raspberry Pi.

3.7 Splitting

- **SynDD**: by driver: **49 training / 10 validation / 10 test** drivers.
- **EBDD**: by driver (2 drivers kept for testing).
- **Open Images** and **Roboflow**: their own published splits.
- **3MDAD** and **DAD**: test only, never used for training.

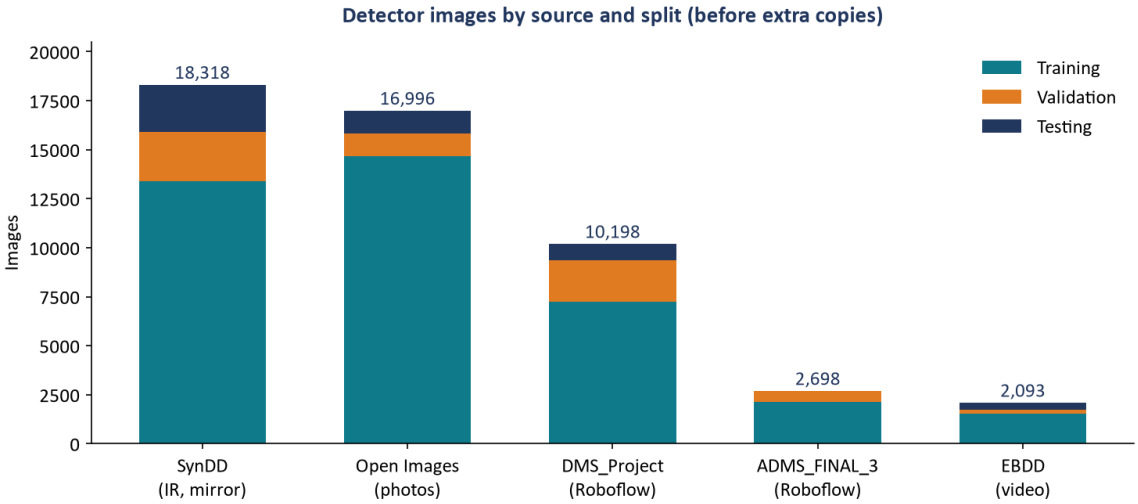


Figure 5: *Detector images by source and split*

> **Fair-testing rule**
> Settings (such as alarm thresholds) are chosen on the **validation drivers only**. The **test drivers are scored once**, at the end.

3.8 Size and class balance

Table 7: *Final detector dataset*

Split	Images	Phone boxes	Drink boxes	Food boxes	No object
Training	53,889	30,193	47,605	24,786	9,327
Validation	6,557	2,099	3,046	1,769	1,514
Test	4,780	1,994	2,528	1,642	1,429

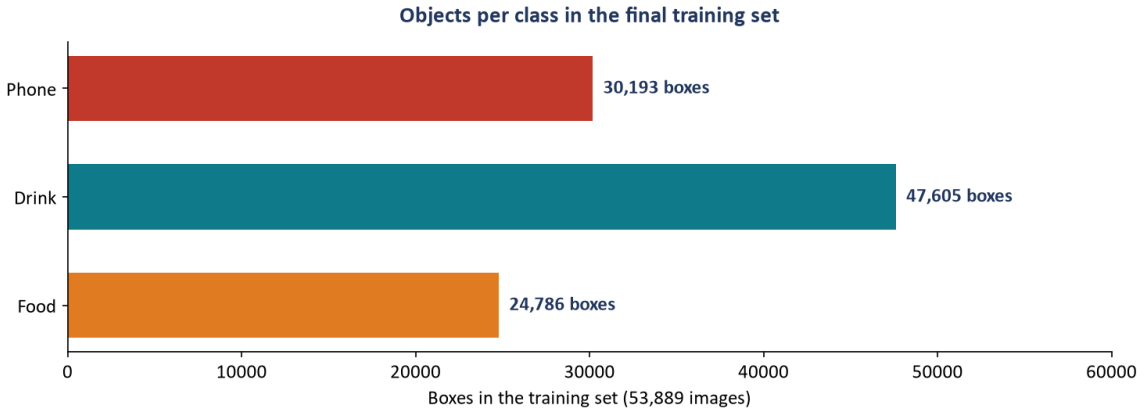


Figure 6: *Objects per class in the training set*

Table 8: *SynDD training frames per action*

Action	Frames	Note
Phone at the ear	4,424	Good amount
Phone in hand (texting)	2,803	Phone often below the camera
Drinking	409	Enough
Eating	3	Weak: food is almost never visible
Normal driving	1,158	No objects
Look-alikes (talking, singing, yawning, hand on head)	4,588	No objects

4. Progress: How It Went

Table 9: *Main steps and results*

#	What we did	Result
1	Version 1: State Farm classifier	Wrong camera angle → replaced
2	First detector, Open Images only	Found the phone in only 1–35% of car frames
3	Downloaded SynDD (69 drivers, infrared, mirror view) and labelled 18,318 frames	Became the main dataset
4	Merged 5 sources (detector v2b)	Phone at the ear found in 72–92% of frames
5	Grey-only training at 512 px + tuned thresholds (v2d)	Test drivers: 80–100% of phone calls caught, 0–5 false alarms per hour
6	Tried 8 more ideas (more views, SAM 2 boxes, fisheye...)	None was better; most added false alarms
7	Tested on 2 unseen datasets (3MDAD, DAD)	Phones still found; many false alarms on a very different camera

Main lesson

Training data must **look like our camera**. Data from other angles **raised false alarms**. Our **own recordings** (Task 7) are now the most important data.

5. Data Quality and Ethics

Table 10: *Quality and ethics checklist*

Check	Status	Details
Test people never seen in training	Done	SynDD, EBDD and MRL split by person; 3MDAD and DAD test only
Thresholds chosen fairly	Done	Validation drivers only
Balanced classes	Partly	Phones boosted with hand copies; food is still rare in car video
Licences respected	Done	Roboflow CC BY 4.0; Open Images CC BY; SynDD, EBDD, DAD research use only
Data kept private	Done	On the project drive only, never on GitHub; EBDD password not in the code
Labels checked by a person	Planned	495 test frames ready to check
Consent for our own recordings	Planned	Signed form before recording

6. Assumptions and Limitations

Assumptions

- The camera sits at the **centre mirror** and uses **infrared** at night.
- The car is **left-hand drive**, so the driver is on the **right** of the image.
- Acted actions (SynDD) look close enough to real ones.

Limitations

- Most video boxes were made by a model and are **not yet hand-checked**.
- **Eating** cannot be learned: food is almost never visible (3 SynDD frames).
- When texting **below the camera**, the phone cannot be seen; a head-down rule helps.
- On a very different camera (DAD, close-up) the **steering wheel** was mistaken for a phone.
- We have **no recordings from our own camera** yet.

7. Next Steps

1. Record **10–20 volunteers** with our camera, day and night (Task 7).
2. Add their **normal driving** to training as "no phone" examples.
3. Hand-check the **495 test frames**.
4. Collect **eating** video where the food is visible, or keep eating out of the alerts.

8. References

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